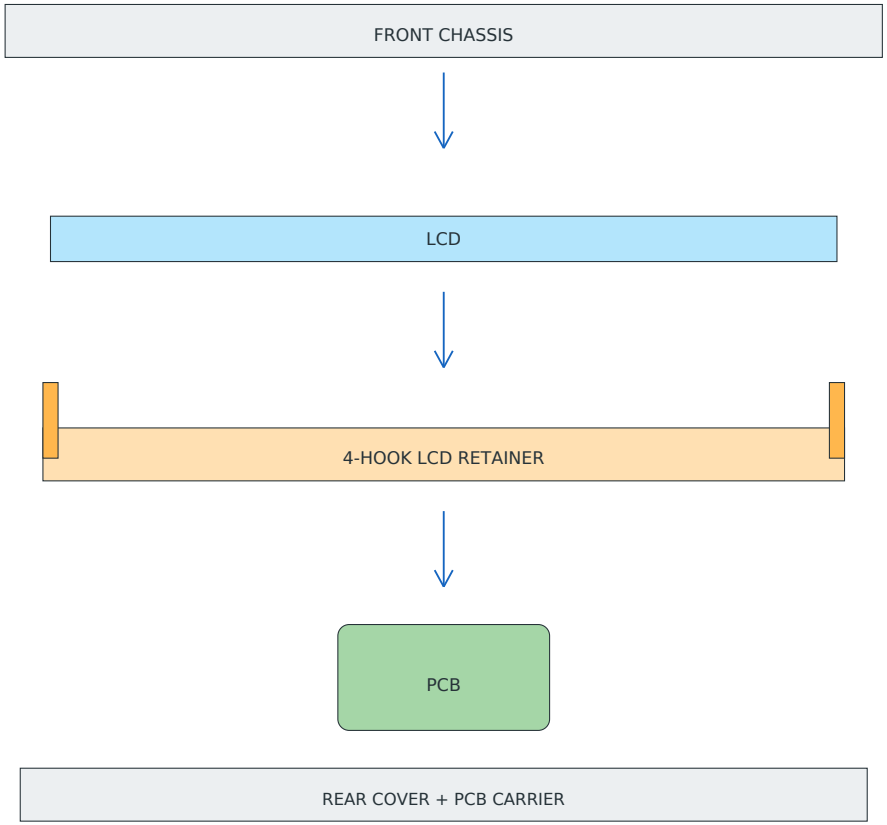
Retention system overview

Each internal item remains retained when the rear cover is removed



REVISION 5 CONTENT

1. LCD retainer snaps directly into chassis at four points.
2. PCB uses a fixed guide on one edge and two flex clips on the other.
3. Two rear-cover connector bulkheads resist insertion loads.
4. Rear cover, PCB carrier, LCD retainer, and panel clips are independent.
5. Rear-cover plate seats on the chassis rear edge; latches match windows.
6. Exact STL-derived A-A to E-E assembly sections are included.
7. HDMI-end two-hole M2 bosses and 20/30 mm deep covers are provided.

ITEM	PRIMARY RETENTION	STATE WITH COVER OFF
LCD	4 retainer hooks	Retained
Tang Nano 9K	fixed guide + 2 clips	Retained on cover
Rear cover	4 chassis latches	Removed as one unit
Panel case	4 panel snap arms	Retained in panel

DESIGN INTENT

No loose internal part is allowed after each assembly stage.

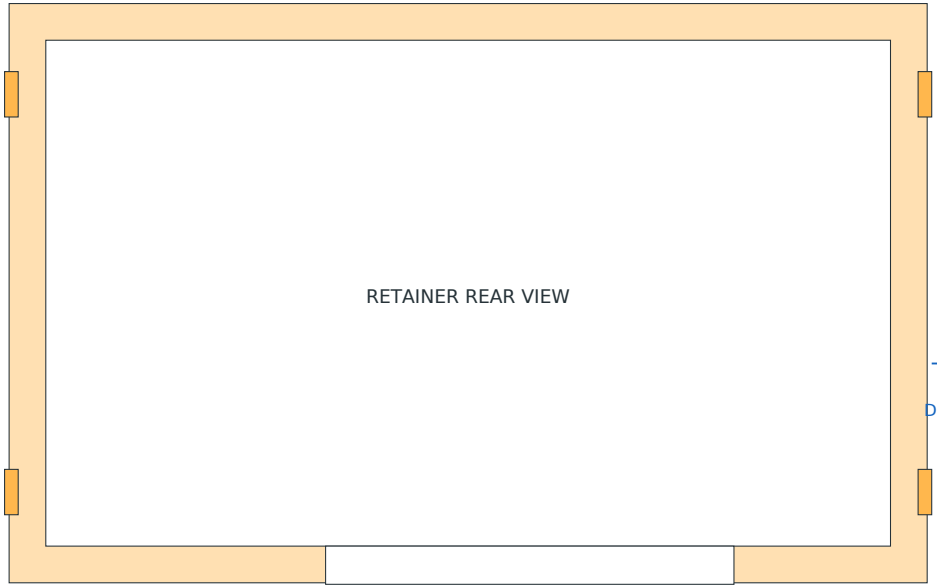
Connector service loads must be reacted by printed stops, not only solder joints.

Snap features are serviceable from the rear without pulling the LCD FPC.

Tang Nano 9K: 70.00 x 26.00 mm | LCD reference: SH500J01Z 120.70 x 75.90 x 3.05 mm

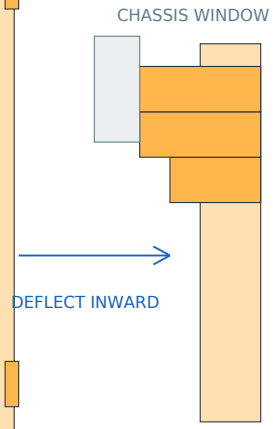
LCD retainer fixation

Four independent cantilever hooks engage chassis side-wall windows



121.40 x 76.60 mm

HOOK SECTION - 4:1 SCHEMATIC



FEATURE	VALUE
Arm thickness	1.20 mm
Arm width	6.00 mm
Overall height	6.50 mm
Free length	about 4.50 mm
Max engagement	0.60 mm
Hook count	4

ASSEMBLY

- Place LCD with the display toward the bezel and FPC toward the relief.
- Push the retainer forward until all four hooks click into the windows.
- Confirm the retainer remains installed without the rear cover.
- To remove, press both hooks on one side inward and lift that edge.

LOAD LIMIT

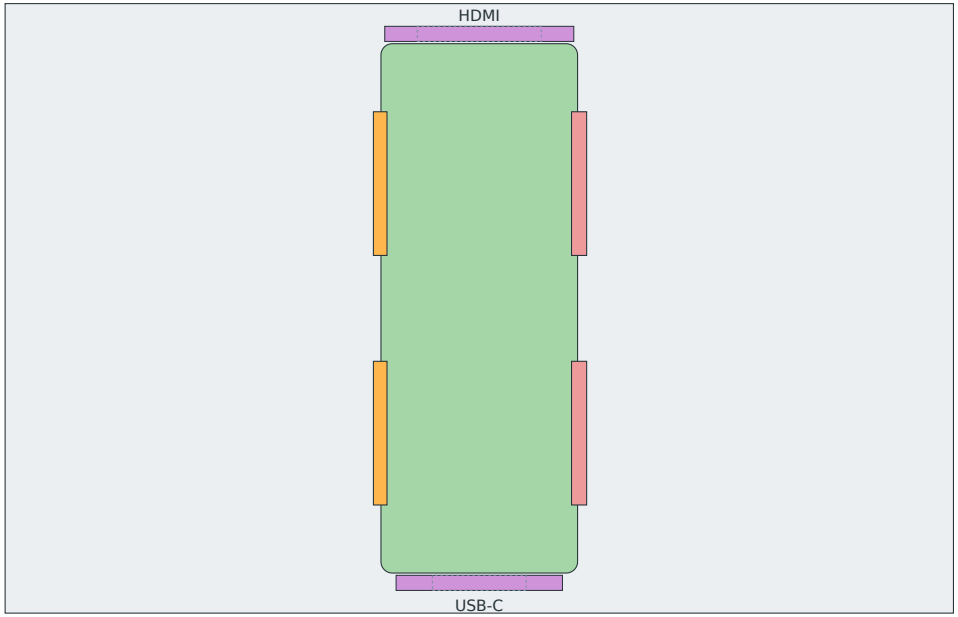
The retainer contacts only the LCD perimeter. Do not preload the active area.

Tang Nano 9K: 70.00 x 26.00 mm | LCD reference: SH500J01Z 120.70 x 75.90 x 3.05 mm

Tang Nano 9K carrier

Fixed-edge insertion, two snap clips, and axial connector-load stops

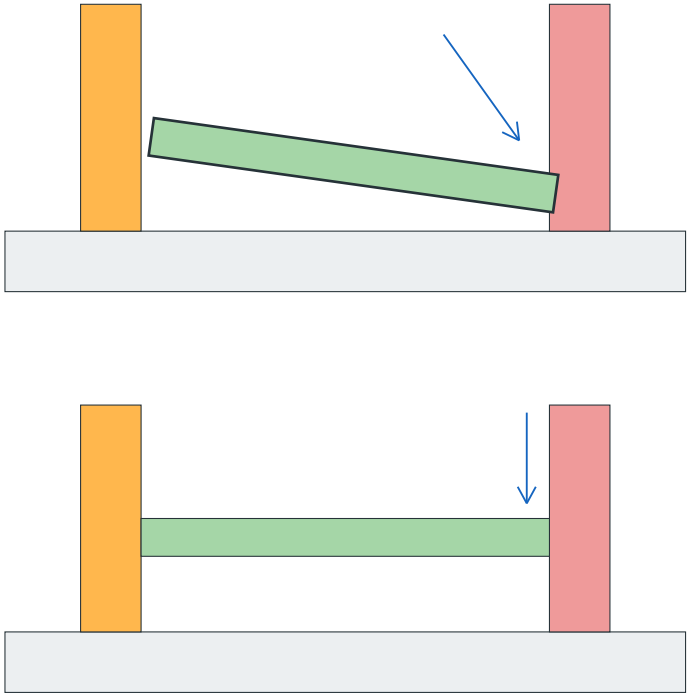
REAR COVER - INBOARD VIEW



Orange: fixed guide | Red: flex clips | Purple: axial stops

FEATURE	VALUE	PURPOSE
Side clearance	0.25 mm/side	FDM fit allowance
Axial clearance	0.30 mm/end	prevents rattle
Support height	7.00 mm	component clearance
Clip thickness	1.20 mm	serviceable flexure
Clip engagement	0.55 mm max	vertical retention
End bulkheads	1 per end	connector load path

INSTALLATION SEQUENCE



1 INSERT FIXED EDGE

2 PRESS SNAP EDGE

REMOVAL

Deflect both red clips outward, lift the right PCB edge, then slide the left edge out from under the fixed lips. Do not lever against the HDMI or USB-C shell.

Tang Nano 9K: 70.00 x 26.00 mm | LCD reference: SH500J01Z 120.70 x 75.90 x 3.05 mm

Assembly and service sequence

Every stage ends with all installed parts positively retained

1 PANEL + LCD

Select the chassis variant matching panel thickness.

Install chassis into the panel cutout.

Place LCD with FPC toward the bottom relief.

2 LOCK LCD RETAINER

Align the retainer FPC relief.

Press until all four side hooks click.

Invert gently: LCD and retainer must remain installed.

3 LOCK PCB TO COVER

Insert left PCB edge under the fixed orange lips.

Press the right edge below both red snap clips.

Confirm both ends sit between the purple end features.

4 CONNECT + CLOSE

Connect the LCD FPC without twisting it.

Orient HDMI top and USB-C bottom.

Engage all four rear-cover latches.

Removal is the reverse order; release clips before pulling.

Tolerance, material, and validation

Prototype verification is required before permanent panel machining

TEST	METHOD	PASS CRITERION	RATIONALE
LCD retention	invert without cover	no release or movement	retainer is independently locked
PCB retention	invert cover only	no release or movement	carrier retains PCB during service
Rattle	manual shake	no impact sound	clearances and stops are effective
USB-C cycling	10 insert/remove cycles	no clip or PCB shift	end stops carry service load
HDMI cycling	10 insert/remove cycles	no clip or PCB shift	end stops carry service load
Cover cycling	5 open/close cycles	all 4 latches retain	serviceability check
FPC inspection	open after cycling	no crease or pinch marks	cable routing remains safe

PRINT BASELINE

Material: PETG preferred | Nozzle: 0.4 mm | Layer: 0.20 mm | Perimeters: 4

PLA is acceptable for a fit prototype but not recommended for repeated snap cycling.

Calibrate elephant-foot and horizontal-hole compensation before changing CAD clearances.

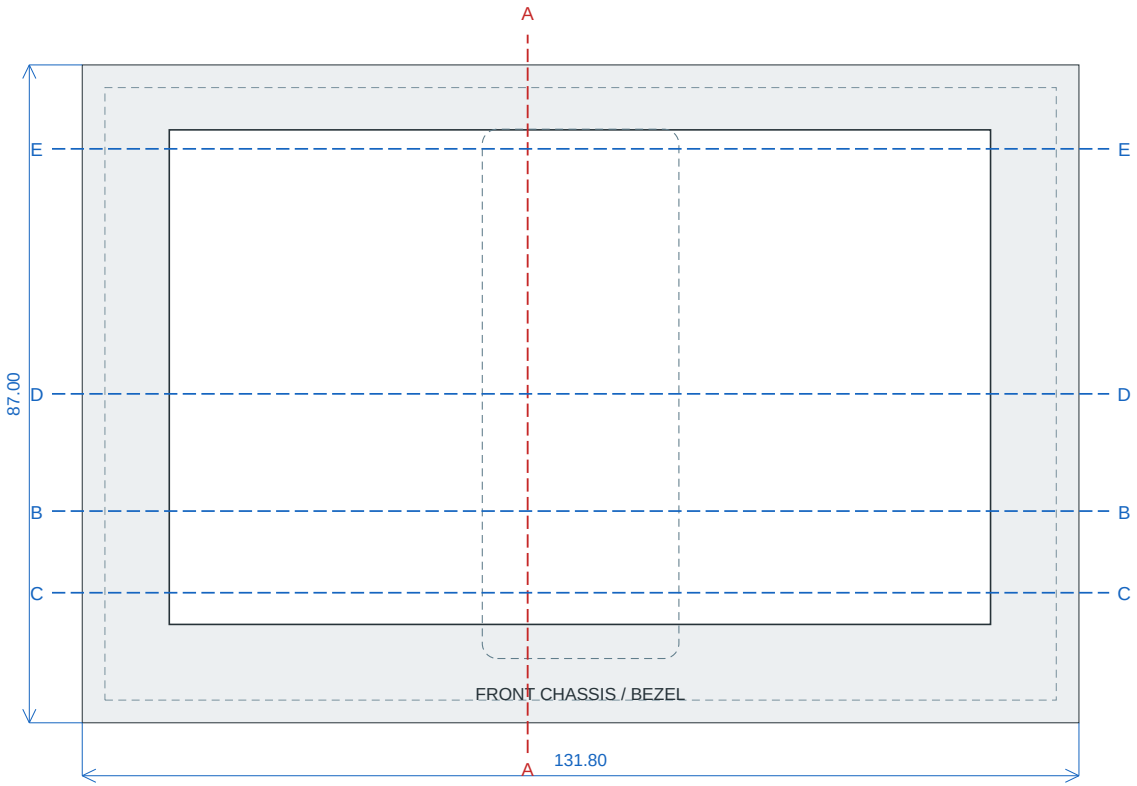
UNVERIFIED HARDWARE VALUES

- USB-C and HDMI shell projection and maximum assembled component height
- LCD FPC tail geometry if the module is not SH500J01Z
- Effective snap force for the user's printer, material, and print orientation
- Cable overmould: USB-C <= 15.0 x 9.0 mm, HDMI <= 23.0 x 12.0 mm

The first print is a fit prototype. Do not machine the final panel before this check.

Exact assembly section index

Five cuts intentionally cross the connector, PCB clip, LCD hook, microSD aperture, and M2 bosses



CUT DEFINITIONS

A-A @ X=58.90

connector openings + PCB end stops + FPC route

B-B @ Y=28.00

PCB support shelves + fixed lip + flex clip

C-C @ Y=17.20

four-hook pair plane + chassis engagement windows

D-D @ Y=43.50

microSD service aperture + rear clearance

E-E @ Y=75.90

two HDMI-end M2 bosses + pilot bores

DRAWING CONTRACT

Filled section cells come from the same rectilinear CSG as the STL meshes.

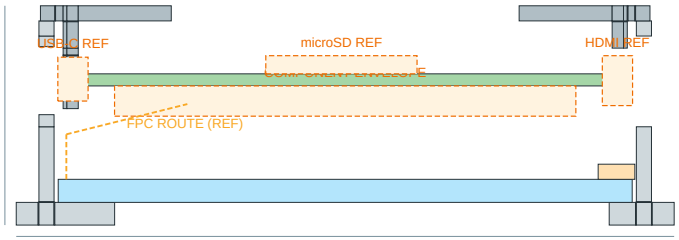
Dashed orange outlines are electronics reference envelopes, not CAD-certified parts.

A blank area can be a deliberate display, latch, connector, or service opening.

Exact sections A-A and B-B

A-A resolves connectors/end stops; B-B proves that the PCB is supported and clipped

A-A Y-Z SCALE 1:1



A-A passes through:

- USB-C and HDMI chassis openings
- one pair of axial PCB end stops
- LCD / retainer / PCB / rear-cover stack
- dashed component and connector reference envelopes
- dashed FPC route corridor

FRONT CHASSIS

LCD VERIFIED BODY

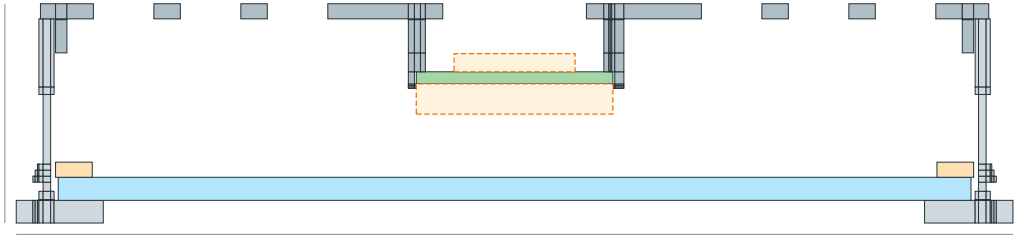
LCD RETAINER

PCB 70 x 26 x 1.6

REAR COVER / CARRIER

REFERENCE ENVELOPE — VERIFY ON HARDWARE

B-B X-Z SCALE 1:1



PCB support condition

The PCB bottom face sits at z=18.40 on four shelves.

The left edge is below a fixed 0.55 mm lip.

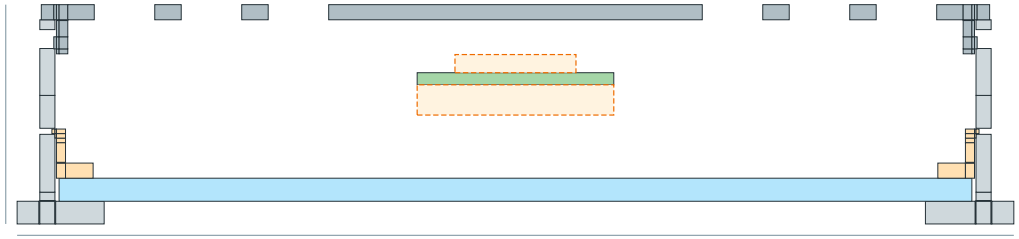
The right edge is retained by two 1.20 mm cantilevers.

Therefore the green PCB section is not suspended in free space.

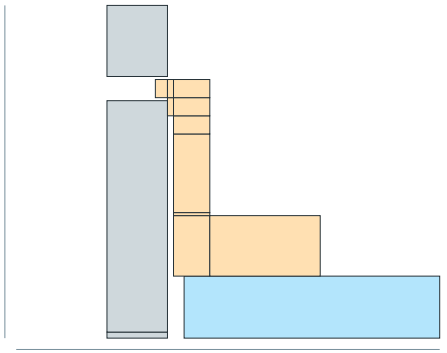
Exact sections C-C and D-D

C-C resolves the retainer hooks; D-D resolves the intentional microSD service opening

C-C X-Z SCALE 1:1



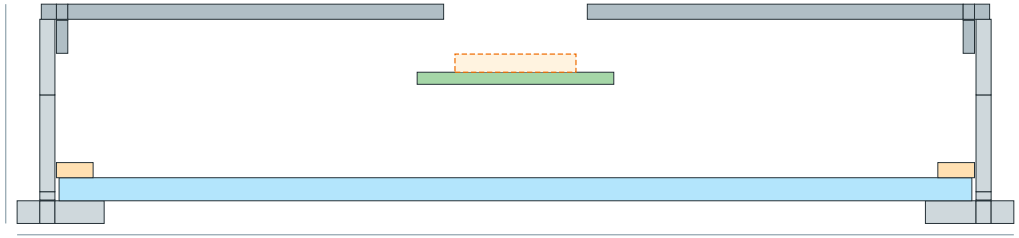
HOOK DETAIL 4:1



Hook head z=10.75..12.55

Window z=11.85..12.65

D-D X-Z SCALE 1:1



The missing rear-cover plate below the PCB is intentional:

it is the 19.00 x 46.00 mm microSD service aperture.

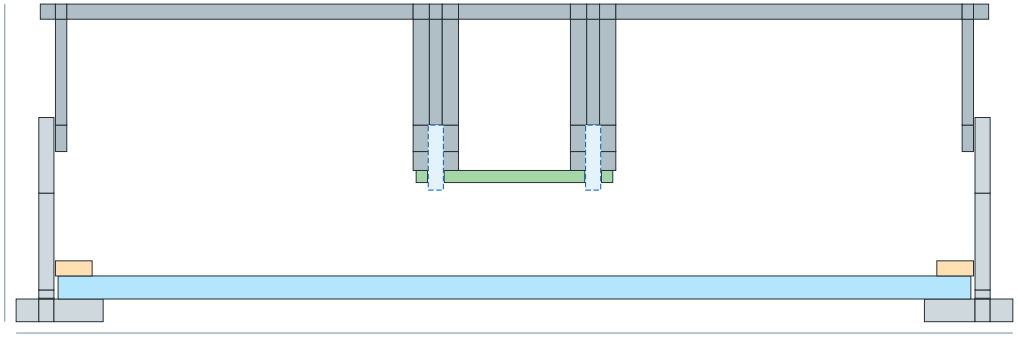
The dashed socket outline is a reference envelope.

Measure the physical socket before production printing.

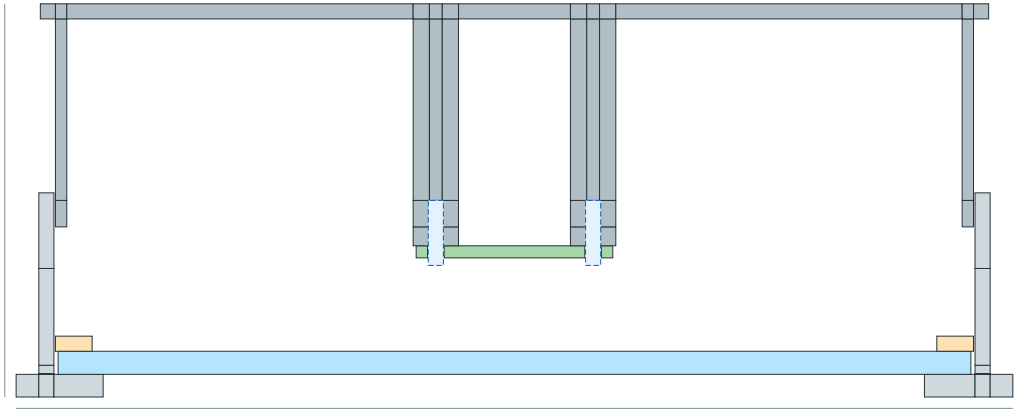
20 mm and 30 mm expansion covers

E-E crosses both HDMI-end screw bosses while PCB and connector planes remain fixed

20 mm REAR CLEARANCE / OVERALL DEPTH 42.00 / SCALE 1:1



30 mm REAR CLEARANCE / OVERALL DEPTH 52.00 / SCALE 1:1



INVARIANT DATUMS

PCB front/rear surfaces remain z=18.40/20.00 mm.

USB-C and HDMI openings remain at the original chassis locations.

Only the rear shell, carrier columns, and cover plate move rearward.

This preserves connector alignment for both expansion variants.

M2 FIXATION

Two bosses align with the two Tang Nano 9K holes at the HDMI end.

The snap lips and clips remain, so screws are optional.

Recommended: M2x8 self-tapping screw, tightened without PCB bowing.

Blue dashed shafts are hardware references; grey features are STL geometry.

EXPANSION HOLD POINTS

Verify daughterboard outline, pin-header height, FPC bend radius,

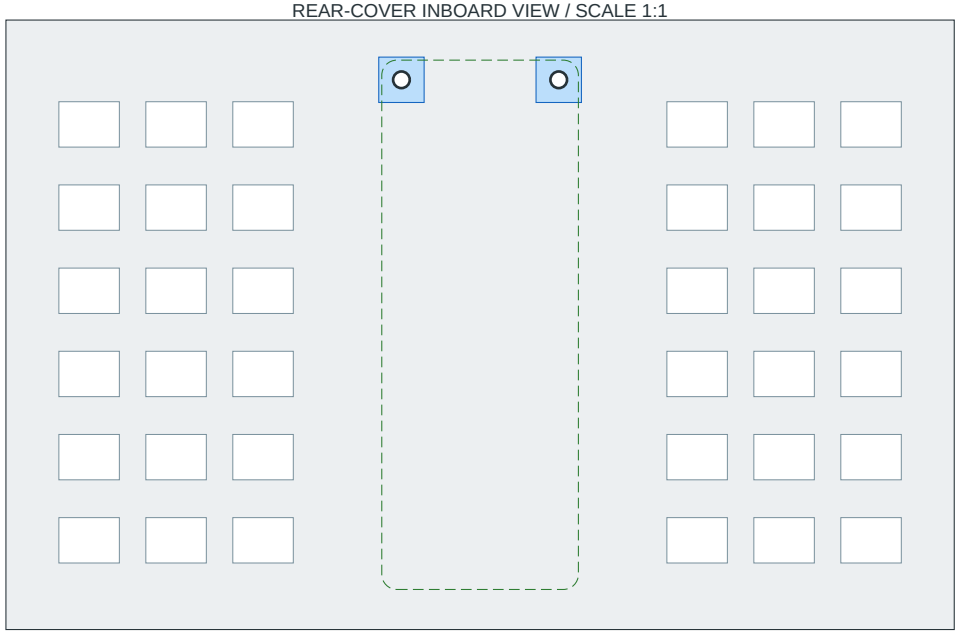
and connector cable bend space before selecting the 20 or 30 mm cover.

Tang Nano 9K: 70.00 x 26.00 mm | LCD reference: SH500J01Z 120.70 x 75.90 x 3.05 mm

Tang Nano 9K two-hole mounting specification

Only the HDMI-end hole pair is used; no USB-C-end mounting holes are assumed

REAR-COVER INBOARD VIEW / SCALE 1:1



Blue: 6 mm bosses | Green dashed: PCB | HDMI end at top

FEATURE	VALUE
PCB mounting holes	2 at HDMI end
Hole diameter ref.	2.20 mm
Hole-centre spacing	20.80 mm
Hole edge offset	2.60 mm
Printed boss	6.00 x 6.00 mm
Pilot bore	1.70 x 1.70 mm
Thread depth	6.00 mm
Screw	M2x8 self-tapping

ASSEMBLY CHECK

1. Seat the PCB under both fixed lips and flex clips.
2. Check both hole centres against the printed pilots.
3. Install both M2 screws evenly; stop before the PCB bends.
4. Confirm the daughterboard does not contact the rear plate.

Do not substitute a longer screw without checking remaining pilot depth.